HWA CHONG INSTITUTION (COLLEGE SECTION)
2016 JC2 H2 BIOLOGY
PRELIMINARY EXAMINTION PAPER 1 & PAPER 3 MARK SCHEME

MULTIPLE CHOICE QUESTIONS

QN	CORRECT ANSWER	QN	CORRECT ANSWER
1	**B**	21	**D**
2	**B**	22	**A**
3	**C**	23	**D**
4	**D**	24	**D**
5	**C**	25	**C**
6	**D**	26	**C**
7	**A**	27	**C**
8	**B**	28	**B**
9	**B**	29	**D**
10	**B**	30	**C**
11	**B**	31	**B**
12	**A**	32	**A**
13	**A**	33	**A**
14	**C**	34	**C**
15	**C**	35	**D**
16	**D**	36	**D**
17	**A**	37	**D**
18	**D**	38	**B**
19	**A**	39	**D**
20	**A**	40	**C**

Question 1

(a) Describe how the restriction enzyme cuts the human DNA and plasmid. [2]

- restriction enzyme cuts at restriction site
- by cleaving phosphodiester bond
- human DNA five restriction sites
- only one restriction site in plasmid

(b)(i) Explain how it is possible to distinguish between bacteria which have taken up a plasmid with human DNA and those which have taken a plasmid without any extra DNA. [4]

- use of sterile velvet to transfer bacteria from master plate of nutrient agar containing ampicillin press velvet onto replica plate of nutrient agar containing tetracycline
- bacteria carrying plasmid with human DNA, the tetracycline resistance gene is disrupted
- bacteria carrying plasmid with human DNA will be killed
- while bacteria with no extra DNA in plasmid will have intact tetracycline resistance gene
- bacteria carrying plasmid with no extra DNA will not killed

(ii) Suggest why genes for antibiotic resistance are now rarely used as markers in genetic engineering. [2]

- risk spread of resistance to other bacteria / ref. to being resistant to antibiotics
- spread of resistance via, conjugation / transformation / incorporation into the genome of bacteria

(c) With reference to Fig. 3.2,

(i) describe the relationship between length of DNA fragment and age of onset of HD. [2]

- the longer the fragment length the earlier the onset
- ref. to shorter fragments moving further / longer fragments moving slower

(ii) explain the difference in positions of DNA fragments from the different members of the family in the test. [3]

- C has recessive, alleles / ref. to homozygote
- A, B and D have one normal and dominant allele / ref. to heterozygotes
- dominant allele gets longer / ref. to increasing CAG repeats from A to B to D

Question 2

a. Identify the type DNA library and outline how the DNA library is produced. [3]

- cDNA library ;;
- Reverse transcriptase catalyses the synthesis of single stranded cDNA using the mature mRNA template of the *dmd* gene;
- DNA polymerase uses the single stranded cDNA as a template to synthesise the complementary/ second DNA strand ;
- Formation of recombinant plasmid/ vector containing the double stranded cDNA and plasmid ;
- The recombinant plasmid/ vector is transformed into bacteria cells ;

b. Explain the limitations of using viruses in the gene transfer of the *dmd* gene. [2]

- There is a limit on the size of the dystrophin gene that can be inserted into a viral vector that results in unsuccessful *dmd* gene expression / AW ;;
- The viral vector may regain virulence and trigger an immune response / AW;;
- There is a possibility of triggering the immune system leading to the host rejecting the *dmd* transgene ;;
- There is random insertion of the *dmd* transgene within the host cells' genome that may induce oncogene activation / AW ;;

c. Describe the unique features of adult mesenchymal stem cells. [2]

- Multipotent stem cells ;
- Mesenchymal stem cells are unspecialised cells ;
- capable of long term self-renewal via mitosis reproducing itself for long periods of time;
- Mesenchymal stem cells can differentiate into specialised cell types under appropriate conditions ;

d. Explain why it is preferable to isolate fibroblast from the the DMD patient. [2]

- Cells are obtained from the same individual and hence ;;
- it will not induce immune rejection when transplanted back into patients ;;

.

e. Suggest the purpose of adding chemical signals (step 4). [1]

1 Differential gene expression to induce the differentiation of iPS to muscle cells ;;

f. Upon transplantation into the patient, suggest two disadvantages of this approach that is similar to conventional gene therapy. [2]

1. Efficiency of transfer of muscle cells into patient maybe low ;;
2. Cells may not produce enough DMD protein ;;
3. The risk of stimulating the immune system leads to rejection by the host ;;

QUESTION 3

(a) **(i)** With reference to Fig. 3.1A, suggest if there is any significant difference between the 2011 and control populations (before 2003) for non-Bt corn. [2]

1. No significance difference between the control and 2011 populations
2. As the error bars overlap with each other

(ii) With reference to Fig. 3.1B, suggest why two types of controls were used. [2]

1. Non-Bt corn is used to determine the effectiveness of Bt protein in decreasing the survival of corn rootworm larvae
2. Control population (before 2003) is used to determine the effectiveness in delaying resistance of Bt protein over the years

(iii) Explain whether expressing one type or two types of Bt proteins is more effective in delaying Bt resistance in corn rootworm.

1. Expressing two types of Bt proteins is more effective
2. In Bt Corn A, survival of larvae increase greatly from 0.075 a.u to 0.275 a.u from control to 2011 populations
3. While in Bt corn B, survival of larvae increase very slightly, almost insignificant**;;**
4. AVP

(b) Explain which type of Bt Corn will best increase crop yield.

1. Corn D
2. Not only can Corn D protect against both the European Corn borer and the corn rootworm;
3. It is also resistant to herbicide

(c) Outline how herbicide tolerant Corn D can be cultured by using calli of Corn C.

1. Cut the region flanking the *bar* gene and the plasmid (pCIB3064) with a restriction enzyme to form a recombinant plasmid
2. Which contains the transgene, the promoter and a genetic marker, such as an antibiotic resistance gene

3. a.The callus culture of Corn C can be treated to produce a protoplast culture
   b. The recombinant plasmid can be introduced to the protoplast culture/ ref. to electroporation/heat shock/gene/ virus-mediated delivery

4. Calli grown in a medium containing the antibiotic
5. Cells who have successfully taken up the recombinant plasmid will be resistant to the antibiotic and will survive, while those who did not take up the recombinant plasmid will die
6. Surviving colonies can be transferred to an agar medium to induce shoot development/ ref. to cytokinin
7. Then transfer to another agar medium after 2-4 weeks to induce root development/ ref. to auxin
8. Ref. to acclimatization of the plantlet

d) Suggest one social issue related to an increase in the use of Corn **D**.

1. After long usage of Corn D, weeds may gain resistance to herbicide
2. Ref. to concerns associated with Bt corn/ pests gain resistance to Bt proteins

Question 4

- Enzyme subtilisin A is a protease that can be used to digest proteins (gelatin) coated on the simulated dirty contact lenses.
- The higher the concentration of subtilisin A in the cleaning solution, higher the rate of gelatin digestion.
- Cut transparent plastic sheet coated with red-coloured gelatin into squares of constant dimensions.
- Place these plastic squares coated with red-coloured gelatin into the specimen vial containing the cleaning solution.
- Start reaction in a thermostatically-controlled water bath for fixed period of time.
- Pour a fixed volume of the cleaning solution into a cuvette.
- Measure the absorbance of the cleaning solution using a colourimeter.
- Carry out a control.
- Perform replicates, repeats and statistical test.
- Calculate the average absorbance of the cleaning solution and obtain rate of gelatin digestion.
- Tabulate data in an appropriate manner.
- Draw sketch of expected trend.

Question 5

(a) Describe the process of polymerase chain reaction. [6]

- DNA denatures when hydrogen bonds are broken, resulting in separation of DNA strands;;
- by heating to 95^0C;;. (accept: 90-100 ^{0}C)

- A pair of primers anneal via complementary base pairing to the 3'ends of the single-stranded DNA template;;
- by cooling to 54 ^{0}C;;. (accept: 50-65 ^{0}C)

- New strands of DNA are synthesised from the position of the primers in the 5' to 3' direction;;
- by *Taq* DNA polymerase;; and
- by heating to 72 ^{0}C;;. (accept: 60-75 ^{0}C)

(b) Explain the advantages and limitations of polymerase chain reaction. [6]

<u>Advantages</u>

- The PCR method is extremely sensitive
- it can amplify sequences from trace amounts of DNA

- PCR is rapid
- A single PCR cycle takes less than 5 minutes / 20 – 30 cycles typically required takes only 2 – 3 hours

- PCR is robust
- PCR can permit amplification of specific sequences from material in which the DNA is badly degraded or embedded in a medium from which conventional DNA isolation is difficult

<u>Limitations</u>

- PCR is prone to contamination
- Due to the extreme sensitivity of PCR, any contamination of the reaction by non-template nucleic acids in the environment could be easily amplified

- PCR may result in infidelity of DNA replication
- *Taq* polymerase used in PCR lacks 3' to 5' exonuclease activity

- Prior knowledge on the DNA sequence is required for the PCR
- Designing specific oligonucleotide primers is required for selective amplification of a particular DNA sequence

(c) Explain the <u>principles</u> of restriction fragment length polymorphism analysis and <u>how it can be used</u> to study genetic variation within a species of organism. [8]

- RFLP are variations in the number / length of restriction fragments generated upon digestion with restriction enzymes;;.
- VNTR resulting in changes in the distance between two restriction sites;;.
- DNA samples of different individuals within a species are isolated and cut/cleaved with the <u>same</u> restriction enzyme for RFLP analysis. ;;
- The digested DNA fragments are subjected to gel electrophoresis, the DNA fragments separate by size ;;.
- Subsequently, Southern blotting was performed to transfer the size-fractionated DNA fragments from the gel onto nitrocellulose membrane ;;.
- Use of radioactively-labelled, single stranded probes complementary to the VNTR / DNA sequence of interest that hybridize to the probes by <u>complementary base-pairing</u> ;;
- Carry out autoradiography to visualise the DNA bands ;;.
- When cut with the same restriction enzyme, genetic variation within a species can be shown by the unique/different DNA fingerprint / banding patterns generated. ;;